
Optimization

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course



ANNOUNCEMENTS



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- ❑ **Midterm is coming up next Tuesday (3 Mar)**
 - worth 20% of total grade
 - Ensure to attend the session you have enrolled in



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- ❑ **Project Proposal Due this Friday (27 Feb)**



TODAY

Optimization Setup



OPTIMIZATION PROBLEM: DEFINITION

Optimization Problem: Determine value of optimization variable within feasible region/set to optimize optimization objective

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{1}$$

- Optimization variable $\mathbf{x} \in \mathbb{R}^n$
- Feasible region/set $\mathcal{F} \subseteq \mathbb{R}^n$
- Optimization objective: $f : \mathcal{F} \rightarrow \mathbb{R}$



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Optimal solution: $\mathbf{x}^* = \underset{\mathbf{x} \in \mathcal{F}}{\operatorname{argmin}} f(\mathbf{x})$



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- Optimization variable $\mathbf{x} \in \mathbb{R}^n$
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- Optimization objective: $f : \mathcal{F} \rightarrow \mathbb{R}$

Optimal solution: $\mathbf{x}^* = \underset{\mathbf{x} \in \mathcal{F}}{\operatorname{argmin}} f(\mathbf{x})$

Optimal objective value $f^* = \min_{\mathbf{x} \in \mathcal{F}} f(\mathbf{x}) = f(\mathbf{x}^*)$



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$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{4}$$

Optimization variable $\mathbf{x} \in \mathbb{R}^n$



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{5}$$

Optimization variable $\mathbf{x} \in \mathbb{R}^n$

- Discrete variables: Combinatorial optimization
- Continuous variables: Continuous optimization
- Mixed: Some variables are discrete, and some are continuous



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{6}$$

Feasible region/set $\mathcal{F} \subseteq \mathbb{R}^n$

- Unconstrained optimization:
- Constrained optimization:



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{7}$$

Feasible region/set $\mathcal{F} \subseteq \mathbb{R}^n$

- Unconstrained optimization: $\mathcal{F} = \mathbb{R}^n$
- Constrained optimization:



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{8}$$

Feasible region/set $\mathcal{F} \subseteq \mathbb{R}^n$

- Unconstrained optimization: $\mathcal{F} = \mathbb{R}^n$
- Constrained optimization: $\mathcal{F} \subset \mathbb{R}^n$



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{9}$$

Feasible region/set $\mathcal{F} \subseteq \mathbb{R}^n$

- Unconstrained optimization: $\mathcal{F} = \mathbb{R}^n$
- Constrained optimization: $\mathcal{F} \subset \mathbb{R}^n$
- Finding a feasible point $\mathbf{x} \in \mathcal{F}$ can already be difficult



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{10}$$



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{11}$$

Optimization objective $f : \mathcal{F} \rightarrow \mathbb{R}$

- $f(\mathbf{x}) = 1$: Feasibility problem



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{12}$$

Optimization objective $f : \mathcal{F} \rightarrow \mathbb{R}$

- $f(\mathbf{x}) = 1$: Feasibility problem
- Simple functions
 - Linear function $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$
 - Convex function (next lecture)



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{13}$$

Optimization objective $f : \mathcal{F} \rightarrow \mathbb{R}$

- $f(\mathbf{x}) = 1$: Feasibility problem
- Simple functions
 - Linear function $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$
 - Convex function (next lecture)
- Complicated functions
 - Can be implicitly represented through an algorithm which takes $\mathbf{x} \in \mathcal{F}$ as input, and outputs a value



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$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{14}$$

Minimization can be converted to maximization (and vice versa)

$$\begin{aligned} \max_{\mathbf{x}} \quad & g(\mathbf{x}) = -f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{15}$$



OPTIMIZATION PROBLEM: DEFINITION

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{16}$$

Minimization can be converted to maximization (and vice versa)

$$\begin{aligned} \max_{\mathbf{x}} \quad & g(\mathbf{x}) = -f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned} \tag{17}$$

Same optimal solution and optimal objective value $g^* = -f^*$



OPTIMIZATION PROBLEM: EXAMPLE

Example: Traveling Salesman Problem (TSP)



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Example: Traveling Salesman Problem (TSP)

- Problem: n cities, distance from city i to city j is $d(i, j)$, find a tour (a closed path that visits every city exactly once) with minimal total distance.
- Variable $\mathbf{x}$: ordered list of cities being visited
 - x_i is the index of the i -th city being visited
- Feasible set $\mathcal{F} = \{\mathbf{x} : \text{“each city visited exactly once”}\}$

$$\mathcal{F} = \{\mathbf{x} : \mathbf{x} \in \{1, \dots, n\}^n; \sum_k \mathbb{I}(x_k = i) = 1, \forall i \in \{1, \dots, n\}\} \quad (20)$$

- Objective function $f(\mathbf{x}) = \text{“total distance when following } \mathbf{x}\text{”}$

$$f(\mathbf{x}) = d(x_n, x_1) + \sum_{k=1}^{n-1} d(x_k, x_{k+1}) \quad (21)$$

OPTIMIZATION PROBLEM: EXAMPLE

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \mathcal{F}\end{array}$$

Example: 8-Queens Problem (Solution 1)



OPTIMIZATION PROBLEM: EXAMPLE

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned}$$

Example: 8-Queens Problem (Solution 1)

- Variable $\mathbf{x}$: location of the queen in each column
 - x_i is the row index of the queen in i -th column
- Feasible set $\mathcal{F} = \{\mathbf{x} : \text{“no queens in the row, col, diag”}\}$

$$\mathcal{F} = \{\mathbf{x} : \mathbf{x} \in \{1, \dots, 8\}^8; x_i \neq x_j, |x_i - x_j| \neq |i - j|, \forall i, j \in \{1, \dots, 8\}\}$$

- Objective function $f(\mathbf{x}) = 1$ (dummy)



OPTIMIZATION PROBLEM: EXAMPLE

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \mathcal{F}\end{array}$$

Example: 8-Queens Problem (Solution 2)



OPTIMIZATION PROBLEM: EXAMPLE

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in \mathcal{F} \end{aligned}$$

Example: 8-Queens Problem (Solution 2)

- Variable x_i, y_i : index of row and column of the i -th queen
- Feasible set $F = \{x, y : \text{“no queens in the row, col, diag”}\}$

$$F = \{x, y : x, y \in \{1, \dots, 8\}^8; \sum_i \mathbb{I}(x_i = k) = 1, \forall k \in \{1, \dots, 8\};$$

$$\sum_i \mathbb{I}(y_i = k) = 1, \forall k \in \{1, \dots, 8\}; |x_i - x_j| \neq |y_i - y_j|, \forall i, j \in \{1, \dots, 8\}$$

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OPTIMIZATION PROBLEM: EXAMPLE

x_i	1.0	2.0	3.0
y_i	2.1	3.98	7.0

Example: Linear Regression



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y_i	2.1	3.98	7.0

Example: Linear Regression

- Problem: Find a such that $y_i \approx ax_i, \forall i = \{1, 2, 3\}$
- Variable a
- Feasible region $\mathbb{R}$
- Objective function $f(a)$?

$$\begin{aligned} \min_a \quad & \sum_{i=1}^3 |y_i - ax_i| \\ \text{s.t.} \quad & a \in \mathbb{R} \end{aligned} \quad (24)$$

$$\begin{aligned} \min_a \quad & \sum_{i=1}^3 (y_i - ax_i)^2 \\ \text{s.t.} \quad & a \in \mathbb{R} \end{aligned} \quad (25)$$



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No general way to solve



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Many algorithms developed for special classes of optimization problems (i.e., when $f(x)$ and $\mathcal{F}$ satisfy certain constraints)



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- Convex optimization problem (CO)



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- Convex optimization problem (CO)
- Linear Program (LP)



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Many algorithms developed for special classes of optimization problems (i.e., when $f(x)$ and $\mathcal{F}$ satisfy certain constraints)

- Convex optimization problem (CO)
- Linear Program (LP)
- (Mixed) Integer Linear Program (MILP)



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- Convex optimization problem (CO)
- Linear Program (LP)
- (Mixed) Integer Linear Program (MILP)
- Quadratic program (QP), (Mixed) Integer Quadratic program (MIQP), Semidefinite program (SDP), Second-order cone program (SOCP), . . .



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- Convex optimization problem (CO)
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Existing solvers and code packages for these problems

- Cplex (LP, MILP, QP), Gurobi (LP, MILP, MIQP), GLPK (LP, MILP), Cvxopt (CO), DSDP5 (SDP), MOSEK (QP, SOCP), Yalmip (SDP), ...



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Why formulate problems as optimization problems?

- For many class of optimization problems, algorithms or algorithmic frameworks have been developed
- Decouple "representation" and "problem solving"



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Lazy mode

- Formulate a problem as an optimization problem
- Identify which class the formulation belongs to
- Call the corresponding solver
- Done !!



Q & A



XKCD

